

HARDWARE EQUIPMENT RECOMMENDATION REPORT

Band01_Rev1 Smartband · Task 3 Preparation

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Current Status: PCB manufactured & available · Firmware compiled (0 errors, 28 May 2026) · No debug probe · No power supply defined

CRITICAL: Power Architecture — Read Before Connecting Anything

⚠ An incorrect power connection will permanently damage the LP5907 LDO regulator on the PCB. AMS1117-3.3 modules **MUST NOT** be used — they will destroy U101.

P1 Header Power Pins — confirmed from Altium schematic:

Pin 12	VBAT	Direct input to LP5907 U101 and U102	✓ CORRECT — connect power here
Pin 17	GND	Ground — connect negative of any power source here	✓ CORRECT — connect GND here
Pin 18	3V3	OUTPUT of LP5907 U101 — do NOT inject power here	✗ WRONG — damages U101 if powered
Pin 16	N/C	Not connected in schematic	✗ Do not use

Why AMS1117-3.3 damages the board: It outputs 3.3V directly into Pin 18 (3V3 net) — the OUTPUT of U101. LP5907 does not permit reverse current on its output. This destroys U101.

DEBUG PROBE RECOMMENDATIONS — 3 Options Ranked

1

RECOMMENDED

SEGGER J-Link EDU Mini

Part: 8.08.91

- JTAG + SWD — CC2640R2F uses JTAG
- EDU license — valid for HSHL academic use
- Compact card form factor + 10-pin ribbon
- Natively supported by CCS
- Standalone flash commander tool included

€68.96 · mouser.de (829-8.08.91)

2

ALTERNATIVE

LAUNCHXL-CC2640R2

TI LaunchPad kit

- Uses onboard XDS110 as debug probe
- Remove isolation jumpers before use
- No CCS config changes needed (XDS110 default)
- Bulkier — full dev board for probe use only
- Useful if CC2640R2F reference needed later

€32.42 · mouser.de (LAUNCHXL-CC2640R2)

3

ALTERNATIVE

SEGGER J-Link BASE

Part: 8.08.28

- Identical function to Option 1 for this project
- Commercial license — not needed for HSHL
- Larger form factor, 20-pin JTAG connector
- No technical advantage over EDU Mini here
- Only if lab policy requires commercial license

€589.05 · mouser.de (829-8.08.28)

JTAG WIRING — P1 Header Connection (Identical for All 3 Probe Options)

Connect probe to P1 header using Dupont M-F jumper wires:

Probe Signal	P1 Pin	Purpose
VTref	Pin 18 (3V3)	Voltage reference sense only — does NOT supply power to board
GND	Pin 17 (GND)	Ground connection
TDI	Pin 11	JTAG Test Data In
TMS	Pin 9	JTAG Test Mode Select
TCK	Pin 7	JTAG Test Clock
TDO	Pin 13	JTAG Test Data Out
nRESET	Pin 15	Target reset line

J-Link CCS config: targetConfigs/CC2640R2F.ccxml → change Connection from 'XDS110' to 'SEGGER J-Link Emulator' → Save. **LaunchPad:** No CCS changes needed.

POWER SUPPLY RECOMMENDATIONS — Connect to P1 Pin 12 (VBAT)

Single-Cell LiPo Battery

1 RECOMMENDED

3.7V · 150mAh · Bare wire leads

Spec: Single-cell LiPo, 3.7V nominal, 150mAh

Connect: Red wire (+) → P1 pin 12 (VBAT)
Black wire (–) → P1 pin 17 (GND)

Why best: Exactly how board was designed. Delivers 73+ hours at 2.032mA avg. Required for final wearable demo.

Lacks: Needs TP4056 charger module between sessions (~€1)

Price: €4.99 · reichelt.de

Adjustable Bench Supply

2 ALTERNATIVE

Set to 3.7V · 200mA+ rated · P1 pin 12

Spec: Adjustable, 3.3–3.8V, ≥200mA current

Connect: Red lead → P1 pin 12 (VBAT) | Set to 3.7V
Black lead → P1 pin 17 (GND) | Current limit: 150mA

Why it works: Simulates LiPo at VBAT. Stable voltage, no drop. Easy power cycling during dev sessions.

Lacks: Board must stay on desk — not suitable for wearable demo. May need buck converter if min output >5V.

When: Initial flashing & first sensor activation — stable voltage more important than portability

⚠ NEVER connect both LiPo and bench supply to VBAT simultaneously — two sources on the same net will conflict and damage the battery and/or LP5907 regulators.

REQUIRED ACCESSORIES & COMPLETE ORDER LIST

Required regardless of which probe or power option is chosen:

Dupont Jumper Wires M-F 20cm (pack of 40) Connect probe to P1 header (7 wires for JTAG) + power supply to VBAT/GND	amazon.de	€8.49
USB-A Cable Connect J-Link EDU Mini or LaunchPad to laptop USB port	Any supplier	€1.20
TP4056 LiPo Charger Module (optional) Recharge LiPo between test sessions — Input: USB 5V, charges single-cell safely	amazon.de	€6.68

Complete Items to Order:

1 — Essential	SEGGER J-Link EDU Mini (8.08.91)	antratek.de / mouser.de	€68.96
2 — Essential	LiPo 3.7V 150mAh bare leads	reichelt.de	€4.99
3 — Essential	Dupont M-F jumper wires (x40)	amazon.de	€8.49
4 — Essential	USB-A cable	Any supplier	€1.20
5 — Recommended	TP4056 LiPo charger module	amazon.de	€6.68

COMPLETE BUILD & FLASH PROCEDURE

✓ Already Completed (28 May 2026)

- 1 Clone repo — TOOLS/build_config.opt and Include/gapbondmgr.h now fixed
- 2 Import & build BLE4 stack library (workspace_v13) from blestack SDK examples
- 3 Copy 3 files (.lib, lib_linker.cmd, ble_r2.symbols) to C:\FlashROM_Library\
- 4 Build smart_band_v001_app in workspace_v12 — 0 errors ✓

→ Still Required (after items arrive)

- 5 Install SEGGER J-Link Software from segger.com (USB drivers + flash commander)
- 6 In CCS: targetConfigs/CC2640R2F.ccxml → Connection → 'SEGGER J-Link Emulator'
- 7 Set supply to 3.7V / connect LiPo → P1 pin 12 (VBAT) + pin 17 (GND)
- 8 Connect probe to P1 header using JTAG wiring table
- 9 In CCS: Run → Debug → firmware flashes → press Play to run
- 10 Verify on smartphone nRF Connect: 'Simple BLE Peripheral' → CHAR1 (0xFFF1) transmitting

SUMMARY — Items to Order

Debug Probe	SEGGER J-Link EDU Mini (Part 8.08.91)	€68.96	Essential
Power	LiPo 3.7V 150mAh bare leads	€4.99	Essential
Wiring	Dupont M-F jumper wires 20cm (×40)	€8.49	Essential
Cable	USB-A cable	€1.20	Essential
Accessory	TP4056 LiPo charger module	€6.68	Recommended

Raspberry Pi: No additional hardware purchase needed — connects wirelessly via BLE Python (bluepy/bleak) to CHAR1 UUID 0xFF1 every 350ms.